

Java Fundamentals & Control Structures - Detailed Lecture Notes

DETAILED COMPREHENSIVE NOTES

Subject: Java Programming | Chapter: Chapter 1: Java Fundamentals & Control Structures | Academic Year: 2026-2027

1. Introduction to the Java Platform & Virtual Machine

Java is a high-level, class-based, object-oriented programming language designed with the philosophy of "Write Once, Run Anywhere" (WORA). This cross-platform portability is achieved through the compilation of source code into an intermediate representation known as bytecode, which is executed by the Java Virtual Machine (JVM).

The JVM architecture comprises three primary subsystems: the Class Loader Subsystem, the Runtime Data Areas, and the Execution Engine. The runtime data areas include the Method Area, Heap, JVM Stacks, Program Counter (PC) Registers, and Native Method Stacks.

- * JDK (Java Development Kit): Contains the compiler (javac), documentation tools, and the JRE necessary for software development.
- * JRE (Java Runtime Environment): Provides libraries, the Java Virtual Machine (JVM), and other components to run Java applications.
- * JVM (Java Virtual Machine): An abstract computing machine that enables a computer to run Java programs through Just-In-Time (JIT) compilation.

2. Primitive Data Types, Variables & Type Conversion

Java defines eight primitive data types: byte (8-bit), short (16-bit), int (32-bit), long (64-bit), float (32-bit IEEE 754), double (64-bit IEEE 754), boolean (true/false), and char (16-bit Unicode UTF-16).

Type casting in Java occurs in two modes: Widening Casting (automatic) when converting a smaller type to a larger type size (e.g., int to double), and Narrowing Casting (manual) requiring explicit casts (e.g., (int) myDouble) which may result in data truncation.

- * Primitive types are stored on the call stack for fast execution, whereas objects are allocated on the garbage-collected heap.
- * Wrapper classes (Integer, Double, Character) enable primitives to be utilized within Java Collections through autoboxing and unboxing.

3. Control Flow Statements & Decision Structures

Decision making statements include if, if-else, nested if, and switch-case. Modern Java (Java 14+) supports enhanced switch expressions with arrow syntax and multiple case labels.

Looping constructs comprise while (entry-controlled), do-while (exit-controlled, executing at least once), standard for, and enhanced for-each loops iterating over arrays and Iterable collections.

- * break: Immediately terminates the innermost loop or switch construct.
- * continue: Skips the remainder of the current iteration and jumps to the loop evaluation step.
- * Labeled breaks and continues allow breaking out of nested outer loops.

4. Methods, Stack Frames & Scope Rules

In Java, all method arguments are passed strictly by value. When passing an object reference, the reference value itself is copied; thus, mutating the object state affects the caller, but reassigning the reference variable does not alter the caller reference.

Method overloading occurs when multiple methods within the same class share the same identifier but have distinct parameter lists (different arity or types).